

Zomato Restaurant Analysis

Business Requirement

Business Scenario

A restaurant discovery and food delivery platform wants to understand customer preferences and restaurant performance across different locations. The company aims to identify the most popular restaurants, top-performing cuisines, preferred restaurant types, and pricing trends to improve customer recommendations and business growth.

As a **Junior Data Analyst**, your responsibility is to analyze the Zomato restaurant dataset using Python and Pandas. You will perform data cleaning, exploratory data analysis (EDA), and visualization to identify popular restaurants, analyze cuisine preferences, study customer ratings, and generate meaningful business recommendations.

Using Python and Pandas, perform data cleaning, exploratory data analysis (EDA), and visualization to answer key business questions and provide actionable recommendations.

Project Objective

Analyze the Zomato restaurant dataset to:

- Identify the most popular restaurants.
 - Analyze customer preferences across different cuisines.
 - Compare restaurant ratings and pricing.
 - Understand restaurant distribution across different areas.
 - Generate business insights and recommendations.
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Dataset Description

The dataset contains information about **restaurants listed on Zomato**, including restaurant name, restaurant type, customer ratings, number of ratings, average cost, online ordering availability, table booking, cuisines served, and restaurant locations.

Dataset Features

Column	Description
Restaurant Name	Name of the Restaurant
Restaurant Type	Type of Restaurant (Cafe, Casual Dining, Quick Bites, etc.)
Rate (Out of 5)	Average Customer Rating
Number of Ratings	Total Customer Ratings
Average Cost (Two People)	Average Cost for Two People
Online Order	Online Ordering Availability
Table Booking	Table Booking Availability
Cuisines Type	Cuisine(s) Served
Area	Restaurant Area
Local Address	Restaurant Locality

Note: Ignore the columns **Unnamed: 0** and **Unnamed: 0.1** during analysis, as they are index columns and do not contribute to the business analysis.

Tasks to Perform

Task 1 – Data Loading

- Import the required libraries.
 - Load the dataset using Pandas.
 - Display the first five records.
 - Display the last five records.
 - Find the number of rows and columns.
 - Display all column names.
 - Check the data types of each column.
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Task 2 – Data Exploration

Perform the following analysis:

- Check for missing values.
 - Check duplicate records.
 - Display summary statistics.
 - Find the number of unique restaurant types.
 - Find the number of unique cuisines.
 - Find the number of unique areas.
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Task 3 – Data Cleaning

Perform the following operations:

- Remove duplicate records (if any).
 - Handle missing values.
 - Remove unnecessary index columns (**Unnamed: 0** and **Unnamed: 0.1**).
 - Ensure **Rate (Out of 5)**, **Number of Ratings**, and **Average Cost (Two People)** have the correct data types.
 - Remove unnecessary spaces from text columns.
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Business Questions

Answer the following questions using Python and Pandas.

Restaurant Analysis

- Which restaurants have received the highest customer ratings?
 - Which restaurants have the highest number of customer ratings?
 - Display the Top 10 restaurants based on customer ratings.
 - Which restaurant types are most common?
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Cuisine Analysis

- Which cuisine is served by the highest number of restaurants?
 - Which cuisines have the highest average customer ratings?
 - Which cuisines receive the highest number of customer ratings?
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Cost Analysis

- What is the average cost for two people?
 - Which are the Top 10 most expensive restaurants?
 - Which restaurant types have the highest average cost?
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Area Analysis

- Which area has the highest number of restaurants?
 - Which areas have the highest average restaurant ratings?
 - Which areas have the most customer engagement based on the number of ratings?
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Restaurant Type Analysis

- Which restaurant type is the most popular?
 - Which restaurant type has the highest average rating?
 - Which restaurant type receives the highest number of customer ratings?
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Online Ordering Analysis

- How many restaurants provide online ordering?
 - Do restaurants offering online ordering receive higher customer ratings?
 - Which restaurant types most frequently offer online ordering?
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Table Booking Analysis

- How many restaurants provide table booking?
 - Do restaurants with table booking have better ratings?
 - Which restaurant types most frequently provide table booking?
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Data Visualization

Create **at least five charts** using Matplotlib.

Mandatory Charts

- Top 10 Restaurant Types (Bar Chart)
- Top 10 Cuisines (Bar Chart)
- Top 10 Areas by Number of Restaurants (Bar Chart)
- Restaurant Rating Distribution (Histogram)
- Online Order Availability (Pie Chart)

Optional Chart

- Top 10 Restaurants by Number of Ratings (Horizontal Bar Chart)
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Business Insights

Write **at least 5 business insights** based on your analysis.

Examples:

- Which cuisines are most popular among customers?
 - Which restaurant types receive the highest ratings?
 - Which areas have the highest restaurant concentration?
 - Does online ordering influence customer ratings?
 - Which restaurants have the strongest customer engagement?
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Business Recommendations

Provide **at least 3 recommendations** for the restaurant discovery platform.

Example recommendations:

- Recommend highly rated restaurants to improve customer satisfaction.
 - Promote popular cuisines in areas with high customer demand.
 - Encourage restaurants to enable online ordering where adoption is low.
 - Highlight restaurants with excellent ratings and strong customer engagement on the platform.
 - Support restaurants in expanding to high-demand areas.
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Submission Requirements

Students must submit:

- Python Notebook (.ipynb)
 - Cleaned Dataset (if modified)
 - Report
 - 3 Business Recommendations
 - 5-minute PowerPoint Presentation
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Evaluation Criteria

Criteria	Marks
Data Loading & Exploration	15
Data Cleaning	15
Data Analysis	30
Visualizations	20
Business Insights	10
Recommendations	10
Total	100